

A K-Means–based Gaussian Process Regression Model for Human Motion Prediction in Human–Robot Collaboration

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Abstract—Human motion prediction plays a critical role in ensuring safe and efficient coordination in human–robot collaboration within manufacturing and service environments. However, human movements exhibit nonlinear and stochastic characteristics, making accurate prediction and reliability assessment challenging. To tackle these issues, the present study introduces a model that combines Gaussian process regression (GPR) with Elbow method-based K-Means, allowing both motion prediction and uncertainty quantification through predictive variance while significantly reducing computational complexity. Results from two-dimensional motion prediction demonstrate that the proposed model achieves reliable prediction accuracy compared with standard GPR while significantly reducing the computational load, thereby decreasing inference time by approximately a factor of eight. Furthermore, the model effectively captures predictive uncertainty, reflecting performance degradation during abrupt changes in human motion.

Index Terms—Gaussian processes, human motion prediction, human–robot collaboration, k-means clustering, uncertainty quantification

I. INTRODUCTION

Human-robot collaboration (HRC) has become increasingly prevalent in modern manufacturing and service industries, where humans and robots cooperate in the same workspace to accomplish shared tasks [1]–[3]. In such collaborative settings, humans contribute cognitive strengths, decision-making, and the flexibility to react to unexpected problems in unstructured environments, whereas robots offer high precision, computational capability, and consistency. The primary objective of HRC systems is to achieve seamless cooperation while ensuring safety, ergonomics, and efficiency [4], [5]. A key challenge in achieving this objective is that robots must proactively anticipate human intentions, rather than merely

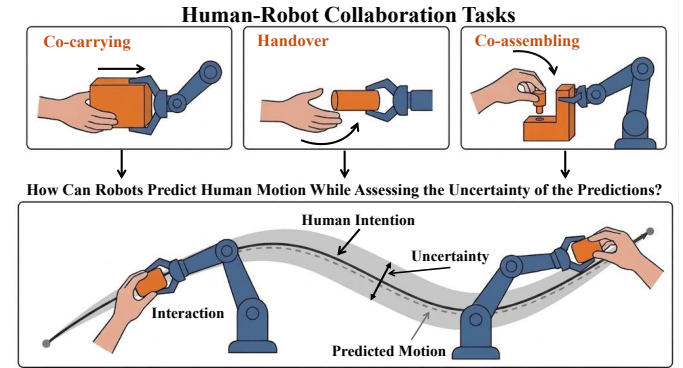


Fig. 1. Illustration of common human-robot collaboration scenarios (co-carrying, handover, and co-assembling) and the corresponding motion prediction pipeline, where the robot infers human intention from observed motion while quantifying predictive uncertainty.

reacting passively to observed motions [6]. Without such capability, robots cannot plan appropriate responses in time, leading to an increased risk of conflicts, reduced task efficiency, and higher human workload. Consequently, as depicted in Fig. 1, accurate human motion prediction is essential for enabling anticipatory motion planning and minimizing collision risks in physical human–robot interaction (pHRI). However, since human motion is inherently nonlinear and stochastic, relying solely on predictions without uncertainty estimates may produce overconfident robot behaviors that are unsafe when unexpected motion changes occur. Therefore, assessing the uncertainty of the predictions enables the robot to make informed decisions and adopt conservative behaviors when confidence is low.

Driven by this challenge, the prediction of human motion has received significant attention in the robotics research community [7]–[9], with existing approaches broadly categorized into model-based and data-driven methodologies. Model-based methods formulate explicit mathematical or physical models to describe and predict human behaviors. For example, the approach in [10] employed an Unscented Kalman filter integrated with an interacting multiple model estimator to predict future human poses based on purely kinematic double integrator models. Taking a different model-based approach, [11] combined a biomechanical model and inverse reinforcement learning to generate explainable motion predictions, integrating them into a model predictive control scheme for safe handovers. However, human movements are inherently nonlinear, stochastic, and highly individual-specific in the event of significant trajectory perturbations or sudden changes in motion patterns. To overcome this limitation, data-driven methods have become increasingly prevalent in recent years [12]–[14]. These approaches learn motion patterns directly from observed data through statistical modeling and machine learning techniques. One of the most widely adopted architectures is recurrent neural networks (RNNs), owing to their ability to capture temporal dependencies in sequential motion data. Yet, standard RNNs suffer from the vanishing gradient problem, limiting their ability to capture long-term dependencies. Advanced variants such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) overcome this limitation, enabling more effective modeling of the nonlinear and stochastic dynamics of human motion. For instance, [13] used a deep LSTM to predict motion in co-carrying tasks, yet prediction errors in practice remain unavoidable due to noise and motion variability. Building on RNN foundations, [14] proposed a residual sequence-to-sequence architecture with a single GRU, effectively resolving first-frame discontinuities while reducing computational cost compared to multi-layer LSTM approaches. Despite these advances, these techniques typically require large training datasets and lack uncertainty quantification mechanisms, which increases risks in pHRI tasks. To mitigate these limitations, Gaussian process regression (GPR) has been proposed to offer a non-parametric Bayesian framework that not only learns effectively from moderate-sized datasets [8], but also provides uncertainty estimates via its predictive variance [15]–[18]. In pHRI, reliable uncertainty quantification is crucial for enabling the robotic system to make informed decisions and adopt conservative behaviors when prediction confidence is low, thereby preventing collisions. However, standard GPR suffers from $\mathcal{O}(N^3)$ time complexity due to the required inversion of the $N \times N$ covariance matrix, where N represents the number of training samples [19]. This computational burden becomes infeasible for real-time applications as the training data expands.

Motivated by the aforementioned analysis, this paper proposes a K-Means-based GPR model (KMGP), in which the Elbow method-based K-Means serves as a Subset of Data (SoD) strategy to select representative samples, significantly

reducing the computational burden in HRC applications. The main contributions of this paper are summarized as follows. First, KMGP achieves prediction accuracy nearly equivalent to standard GPR while reducing inference time by approximately eight times using only 20% of the original training samples. Second, two representative HRC scenarios are considered to demonstrate the KMGP’s uncertainty quantification capability and prediction performance compared to standard GPR. Furthermore, a theoretical analysis is provided to explain the behavioral differences between KMGP and standard GPR under abrupt motion changes.

The remainder of this paper is organized as follows. The problem of human motion prediction is formulated as a regression task in Section II. Section III presents the GPR framework integrated with the Elbow method-based K-Means for computational efficiency. Section IV provides validation setup and performance evaluation across two specific scenarios. Finally, the conclusions and future works are presented in Section V.

II. PROBLEM FORMULATION

The primary objective of this work is to predict human hand motion $\mathbf{p}_t^\top \in \mathbb{R}^d$. At each time step t , given an observation window $\mathbf{X}_t = [\mathbf{p}_{t-h+1}^\top, \dots, \mathbf{p}_t^\top] \in \mathbb{R}^{dh}$ encompassing h historical time steps, the goal is to predict future motions. However, since human motion is inherently stochastic and nonlinear, predictions carry unavoidable uncertainty, and long-horizon forecasting is prone to error accumulation [15], [18]. Furthermore, to ensure safety and efficiency in pHRI, accurate motion prediction must be coupled with real-time uncertainty quantification at a reasonable computational cost. Consequently, this work focuses on short-term prediction, where the predictive output is defined as $\mathbf{Y}_t = \mathbf{p}_{t+1}^\top \in \mathbb{R}^d$.

III. METHOD

A. Architecture Overview

To formulate the multi-dimensional prediction from a historical window to the future motion, the problem is decomposed

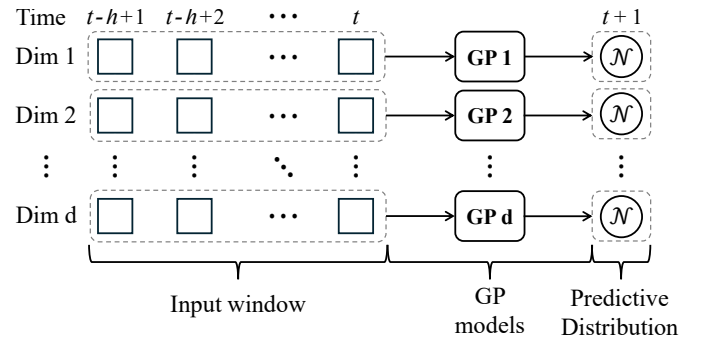


Fig. 2. Multi-dimensional GP architecture. The input observation window encompasses h past time steps and is decomposed into d separate dimensions. Each dimension is then modeled by an independent GP. Finally, each GP processes the h -step window to predict the future state, producing its own Gaussian predictive distribution.

across dimensions, where each dimension is modeled by an independent GP as shown in Fig. 2.

Formally, for a given dimension, let $\mathbf{x}_i \in \mathbb{R}^h$ represent the historical observations, and y_i is the corresponding future state. The relationship between the past observations and the future state is modeled as a latent function $f: \mathbb{R}^h \rightarrow \mathbb{R}$ with additive Gaussian noise:

$$y_i = f(\mathbf{x}_i) + \epsilon, \quad (1)$$

where $\epsilon \sim \mathcal{N}(0, \sigma_n^2)$ and σ_n^2 denotes the noise variance.

B. Gaussian Process Regression

GP is used to define a distribution over functions, where any finite collection of function values has a joint Gaussian distribution [17]. A GP is fully specified by its mean function and covariance function:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')), \quad (2)$$

in which $m(\mathbf{x})$ and $k(\mathbf{x}, \mathbf{x}')$ are the mean function and the covariance function, respectively. The mean function is normally set to zero, and a widely used covariance function is the squared exponential (SE) with automatic relevance determination (ARD) [17], defined as

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{x}')^\top \mathbf{L}^{-1}(\mathbf{x} - \mathbf{x}')\right), \quad (3)$$

where σ_f^2 is the signal variance, and $\mathbf{L} = \text{diag}[l_1^2, l_2^2, \dots, l_h^2]$ comprises the length scales across the h input features.

Given a training dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ containing N training samples for any given dimension, and the noisy observation model in (1), the set of hyperparameters $\theta = \{\sigma_f^2, \mathbf{L}, \sigma_n^2\}$ is optimized by maximizing the log-marginal likelihood, given by

$$\log p(\mathbf{y}|\mathbf{X}, \theta) = -\frac{1}{2}\mathbf{y}^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} - \frac{1}{2} \log |\mathbf{K} + \sigma_n^2 \mathbf{I}| - \frac{N}{2} \log 2\pi, \quad (4)$$

where $\mathbf{y} = [y_1, y_2, \dots, y_N]^\top$ is the vector of training outputs, and

$$\mathbf{K} = \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}_1) & k(\mathbf{x}_1, \mathbf{x}_2) & \cdots & k(\mathbf{x}_1, \mathbf{x}_N) \\ k(\mathbf{x}_2, \mathbf{x}_1) & k(\mathbf{x}_2, \mathbf{x}_2) & \cdots & k(\mathbf{x}_2, \mathbf{x}_N) \\ \vdots & \vdots & \ddots & \vdots \\ k(\mathbf{x}_N, \mathbf{x}_1) & k(\mathbf{x}_N, \mathbf{x}_2) & \cdots & k(\mathbf{x}_N, \mathbf{x}_N) \end{bmatrix} \quad (5)$$

is the $N \times N$ covariance matrix of the training data.

For a new test input $\mathbf{x}_*$, the predictive distribution of the future state y_* is given by $p(y_*|\mathcal{D}, \mathbf{x}_*) = \mathcal{N}(\mu_{y_*}, \Sigma_{y_*})$, with its mean

$$\mu_{y_*} = \mathbf{K}_* (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y}, \quad (6)$$

and variance

$$\Sigma_{y_*} = k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{K}_* (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{K}_*^\top, \quad (7)$$

where $k(\mathbf{x}_*, \mathbf{x}_*)$ denotes the covariance of the test input with itself, and $\mathbf{K}_*$ is the covariance vector between the test input $\mathbf{x}_*$ and the training inputs:

$$\mathbf{K}_* = [k(\mathbf{x}_*, \mathbf{x}_1) \quad k(\mathbf{x}_*, \mathbf{x}_2) \quad \cdots \quad k(\mathbf{x}_*, \mathbf{x}_N)]. \quad (8)$$

As depicted in Fig. 2, the test input $\mathbf{X}_* \in \mathbb{R}^{dh}$ is decomposed into d separate dimensions, each independently processed by a GP to yield a Gaussian predictive distribution. Since the predicted motion $\mathbf{Y}_* \in \mathbb{R}^d$ comprises d independent Gaussian components, its joint distribution follows a multivariate Gaussian $\mathcal{N}(\boldsymbol{\mu}_{\mathbf{Y}_*}, \boldsymbol{\Sigma}_{\mathbf{Y}_*})$ with the mean vector

$$\boldsymbol{\mu}_{\mathbf{Y}_*} = [\mu_{y_{1*}}, \mu_{y_{2*}}, \dots, \mu_{y_{d*}}]^\top, \quad (9)$$

and the covariance matrix

$$\boldsymbol{\Sigma}_{\mathbf{Y}_*} = \text{diag}[\Sigma_{y_{1*}}, \Sigma_{y_{2*}}, \dots, \Sigma_{y_{d*}}]. \quad (10)$$

While this formulation provides a principled probabilistic framework for multi-dimensional prediction, its real-time deployment is limited by computational cost. As seen in (6) and (7), standard GPR suffers from a cubic time complexity of $\mathcal{O}(N^3)$ due to the required inversion of the $N \times N$ covariance matrix $\mathbf{K} + \sigma_n^2 \mathbf{I}$. This computational burden severely limits the scalability of the GP, making it unaffordable for real-time motion prediction in continuous HRC tasks. To address this limitation, the SoD approximation serves as one of the most fundamental global approximation strategies to scale up the standard GP [19]. By constructing the model on a subset of size M ($M \ll N$), the SoD method effectively reduces the time complexity to $\mathcal{O}(M^3)$. However, the effectiveness of the SoD approach depends critically on the selection method and the subset size, which must be carefully chosen to preserve the characteristics of the training dataset. To fulfill this requirement, this work proposes a K-Means-based subset selection strategy.

C. Elbow method-based K-Means

In this study, the Elbow method-based K-Means is employed to select a representative SoD from the full training dataset. Given a predefined number of clusters K , the algorithm partitions the input set $\{\mathbf{X}_1, \dots, \mathbf{X}_N\}$ into K clusters $\{S_1, \dots, S_K\}$ through the following iterative steps:

- 1) *Initialization*: Select K data points randomly from the training set to serve as the initial centroids.
- 2) *Cluster Assignment*: For each data point $\mathbf{X}_i$, calculate the Euclidean distance to each centroid and assign it to the nearest cluster.
- 3) *Centroid Update*: Compute the mean of all data points assigned to each cluster S_k to update its centroid $\mathbf{C}_k$.
- 4) *Iteration*: Repeat steps (2) and (3) until the cluster assignments no longer change.

A standard measure of partition quality is the sum of squared errors (SSE), given by

$$\text{SSE} = \sum_{k=1}^K \sum_{\mathbf{X}_i \in S_k} \|\mathbf{X}_i - \mathbf{C}_k\|^2. \quad (11)$$

This measure is then used by the Elbow method to determine the optimal number of clusters, denoted M , by evaluating the SSE in (11) across a range of K values [20]. On the resulting curve, as K increases, the SSE initially declines rapidly. However, beyond a certain point, the SSE continues to decline but at a significantly slower rate, forming a distinct

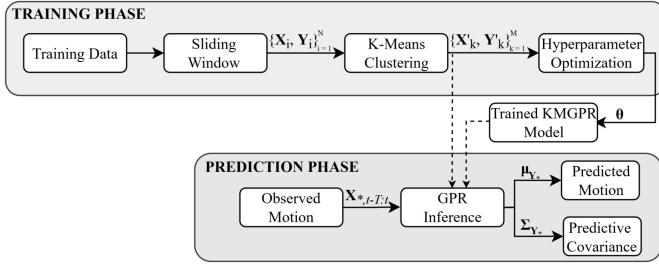


Fig. 3. The proposed framework for KMGPR consists of an offline training phase and an online prediction phase. In the offline training phase, K-Means clustering selects a representative SoD $\{(\mathbf{X}'_k, \mathbf{Y}'_k)\}_{k=1}^M$ from the full dataset for hyperparameter optimization. During the online prediction phase, the trained KMGPR model outputs the predicted position $\mu_{\mathbf{Y}_*}$ and the predictive covariance $\Sigma_{\mathbf{Y}_*}$ based on the observed motion window $\mathbf{X}_{*,t-T:t}$.

elbow point. The value of K at this elbow point is identified as the optimal M .

Once the optimal number of clusters M is identified, the representative SoD $\{(\mathbf{X}'_k, \mathbf{Y}'_k)\}_{k=1}^M$ is obtained as follows. Within each cluster S_k , the training input $\mathbf{X}_i \in S_k$ closest to the centroid $\mathbf{C}_k$ is extracted. This input, together with its corresponding output $\mathbf{Y}_i$, constitutes the representative pair $(\mathbf{X}'_k, \mathbf{Y}'_k)$. Replacing the full training dataset with this representative SoD reduces the overall time complexity from $\mathcal{O}(N^3)$ to $\mathcal{O}(M^3)$. The framework of the proposed KMGPR model is illustrated in Fig. 3.

IV. VALIDATION EXAMPLE

This section presents a comparative evaluation of the proposed KMGPR model against the standard GPR.

A. Validation Setup

The two-dimensional motion data used for training and testing were generated using MATLAB. All validations were conducted on an Intel Core i5-12400 CPU with 16 GB of RAM. The detailed parameter settings for KMGPR and GPR are summarized in Table I.

1) *Preparation Stage*: The training dataset consists of 30 trajectories representing target-directed human motions. All trajectories originated from the initial position (0, 0) and navigated towards three distinct targets within 10 seconds at a sampling interval of 0.1 s. According to the validation split in Table I, 24 trajectories were allocated for the training phase, while the remaining 6 were used to validate the training process.

TABLE I
PARAMETERS FOR MODEL TRAINING

Parameters	Values
Window size	10
Kernel function	SE
Epochs	250
Initial learning rate	0.05
Validation split	0.2
Optimizer	Adam
Loss function	Mean square error

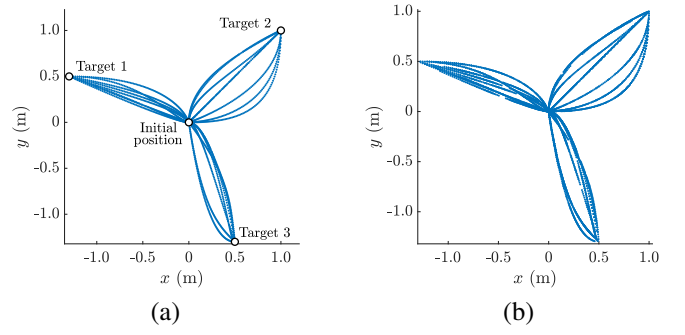


Fig. 4. Distribution of the training data: (a) the full dataset ($N = 2424$ windows), and (b) the SoD generated by K-Means clustering ($M = 484$ windows).

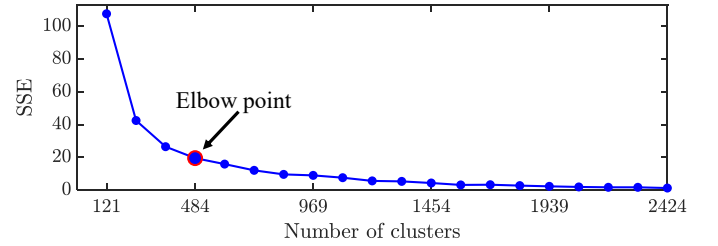


Fig. 5. The SSE curve using the Elbow method for determining the optimal number of clusters. The subset size was evaluated from 121 to 2424 (5% to 100% of the total windows) to ensure sufficient data for model training.

The 24 training trajectories were processed through a sliding window approach, as depicted in Fig.4 (a), generating a total of 2424 windows, which imposes a significant computational burden on the standard GPR.

To reduce this computational burden, the Elbow method-based K-Mean algorithm in Section III-C was applied to extract a SoD from the training data. By evaluating the SSE curve via the Elbow method in Fig. 5, the optimal number of clusters was determined to be $M = 484$. This corresponds to an 80% reduction in the size of training data (from 2424 to 484 windows). As can be seen from Fig. 4(b), despite retaining only 20% of the original windows, the SoD effectively preserves the coverage of the full dataset, demonstrating the robustness of K-Means in extracting a representative subset.

2) *Validation Scenarios*: To evaluate the predictive performance of the proposed model, two distinct test scenarios were designed to reflect typical situations encountered in HRC.

The first scenario focuses on nominal tasks to evaluate the model's accuracy across trajectories that are consistent with the training data. In this scenario, the human hand moves from the initial position (0, 0) to one of three predefined targets.

The second scenario evaluates the model's robustness when facing motion patterns not seen during training. In this case, the hand initially moves toward Target 2 at a normal speed. However, the human operator suddenly stops, remains stationary for several seconds, and eventually resumes moving toward the target.

TABLE II
PERFORMANCE COMPARISON ON NOMINAL TRAJECTORIES

Model	MSE ($\times 10^{-6} \text{m}^2$)				Prediction Time (s)
	Target 1	Target 2	Target 3	Overall	
KMGPR	0.12	0.40	0.12	0.21	4.62
GPR	0.09	0.36	0.10	0.19	41.09

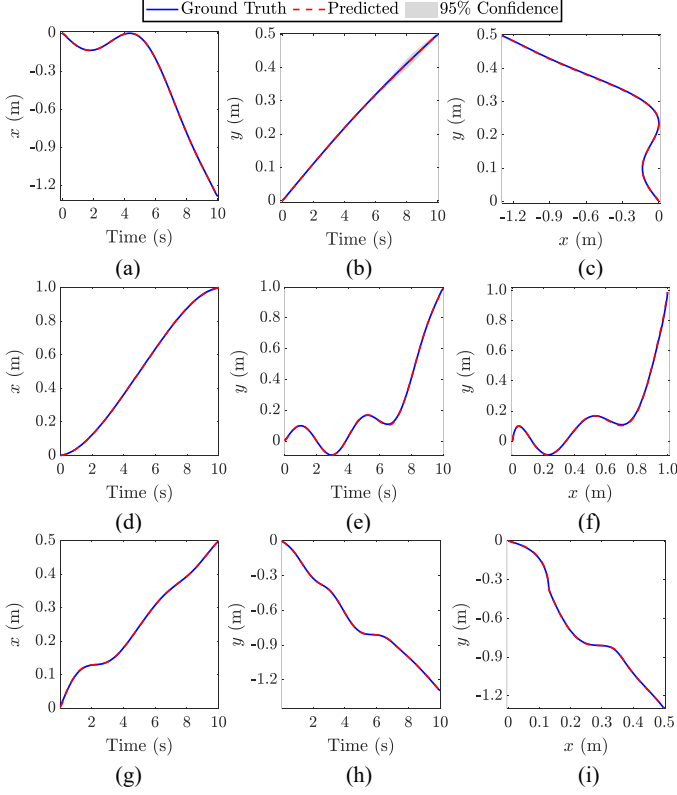


Fig. 6. Prediction performance of KMGPR with uncertainty estimates shown as 95% confidence intervals under nominal trajectories: (a, d, g) are x -axis position; (b, e, h) are y -axis position; (c, f, i) are two-dimensional trajectories for Target 1, Target 2, and Target 3, respectively.

B. Validation Results and Analytics

1) *Scenario 1 - Nominal Trajectories*: In this scenario, both models including KMGPR and GPR are evaluated on nominal trajectories toward three learned targets. The quantitative results in Table II show that KMGPR achieves an overall MSE of $0.21 \times 10^{-6} \text{m}^2$, while standard GPR achieves $0.19 \times 10^{-6} \text{m}^2$, demonstrating that both models maintain nearly equivalent prediction accuracy across all three targets. Regarding computational efficiency, KMGPR reduces the prediction time by approximately a factor of eight compared to standard GPR. This is attributed to the elbow method-based K-Means, which extracts a SoD of $M = 484$ samples from the initial training set of $N = 2424$, which is then utilized in the prediction phase as illustrated in Fig. 3.

The uncertainty estimation capability of the KMGPR model is evaluated using the 95% confidence interval. As depicted in Fig. 6, despite relying on merely 20% of the original data, the

KMGPR model maintains consistently low uncertainty across three predefined targets, demonstrating that the extracted SoD still provides sufficient data to yield high-confidence predictions in nominal scenarios.

The results of Scenario 1 confirm that applying K-Means to select a representative subset still maintains prediction accuracy nearly equivalent to standard GPR, while preserving uncertainty estimation capability and reducing significant computational workload.

2) *Scenario 2 - Abrupt Motion Changes*: In Scenario 2, both models are evaluated on an atypical trajectory with abrupt changes in motion. Specifically, the trajectory consists of two distinct phases: the human operator first accelerates from $t = 3.3$ s, then stops abruptly at $t = 4.0$ s. When evaluated over the entire trajectory, standard GPR achieves a slightly lower overall MSE than KMGPR ($4.42 \times 10^{-6} \text{m}^2$ versus $5.23 \times 10^{-6} \text{m}^2$). However, this metric does not fully capture the behavioral differences observed across distinct phases of this scenario.

As the human operator begins accelerating around $t = 3.3$ s, the predictions exhibit inherent latency, failing to track the motion instantaneously and thus falling behind the ground truth. Subsequently, as the motion decelerates and stops abruptly at $t = 4.0$ s, both models overshoot the stationary position. Notably, the uncertainty estimates of both models expand significantly at this stage, indicating a sharp drop in prediction confidence because such sudden deceleration patterns are absent from the training data. From this point, the two models exhibit markedly different behaviors. As shown in Fig. 7(c), after overshooting, KMGPR returns to the stationary position along a stable recovery trajectory without chattering. In contrast, Fig. 7(f) shows that standard GPR exhibits a highly oscillatory trajectory, making a wider deviation before eventually converging to the stationary position. This difference

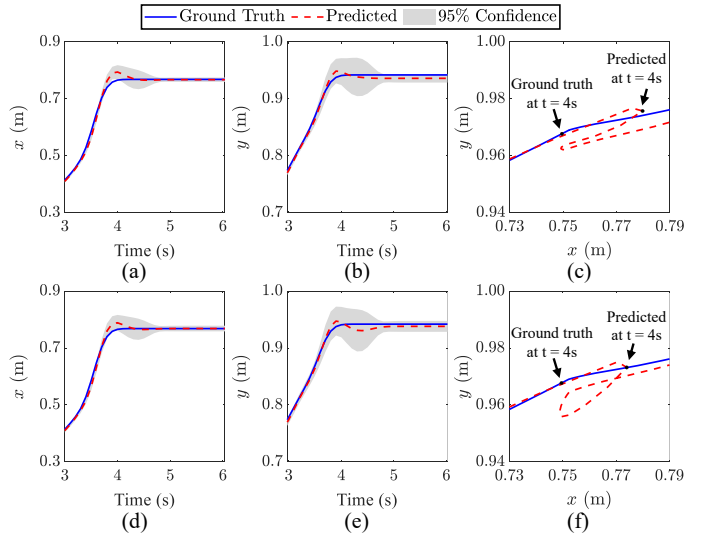


Fig. 7. Comparing two models under abrupt motion changes with uncertainty estimates shown as 95% confidence intervals: (a), (d) are x -axis position; (b), (e) are y -axis position; (c), (f) are two-dimensional trajectories of the KMGPR and the standard GPR, respectively.

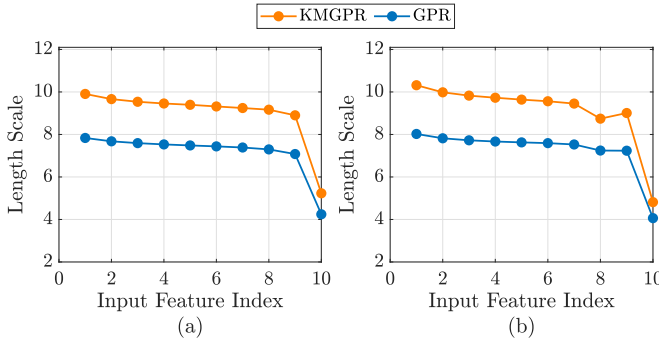


Fig. 8. Comparison of length scales across input features for standard GPR and KMGPR on (a) x -axis, (b) y -axis

is attributed to the ARD length-scale values learned by each model, as presented in Fig. 8. Across all input features, KMGPR learns larger length scales than standard GPR. These length-scale values are the direct consequence of the log-marginal likelihood optimization, in which the balance between the data-fit term $(-\frac{1}{2}\mathbf{y}^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y})$ and the complexity penalty $(-\frac{1}{2} \log |\mathbf{K} + \sigma_n^2 \mathbf{I}|)$ in (4) is governed by the size of the training dataset [17]. For the standard GPR, the large training dataset places greater pressure on the data-fit term, driving the algorithm to converge toward smaller length scales. In contrast, K-Means clustering significantly reduces the number of training points, thereby diminishing the pressure on the data-fit term and allowing the optimization to converge at larger length scales. Under such conditions, the small length scales of standard GPR produce sharp variations in the predictive function [17]. This causes the predictions to oscillate strongly, deviating significantly from the human's actual stopping intention. Unlike standard GPR, with larger length scales, KMGPR interprets this unseen state through a slowly varying function, maintaining predictions that gradually converge to the stationary state.

V. CONCLUSION AND FUTURE WORK

This paper proposed a KMGPR model for the human motion prediction problem in HRC. By utilizing K-Means clustering to select a representative subset, the proposed method significantly reduces computational complexity compared to standard GPR, while maintaining motion prediction accuracy and uncertainty quantification through predictive variance. Validation results demonstrate that the model maintains prediction accuracy nearly equivalent to standard GPR while reducing computation time by approximately eight times. In the abrupt motion change scenario, KMGPR exhibits greater prediction stability with significantly fewer oscillations compared to standard GPR. Future work will focus on validating with real-world pHRI in three-dimensional framework and developing adaptive schemes based on predicted uncertainty.

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